



## Shubhendra Pratap Singh

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### PROFESSIONAL SUMMARY

Mechanical Engineer with 8+ years of experience in R&D, product design, rapid prototyping, and mechanical systems development. Expertise in developing new product concepts, verifying specifications, designing fixtures, testing processes, algorithms and process refinement for LLM-assisted bio-inspired design. Proficient in SolidWorks, MATLAB, Python, and MS Office for technical documentation and process automation. Delivered cost-saving automation tools and contributed to bio-inspired designs, with publications in SAE/ASME.

### PROFESSIONAL EXPERIENCE

#### **Project Engineer**

FEV India Pvt. Ltd. (Automotive Research Company), Pune, India

Nov 2017 – Apr 2021

- Developed macro-based automation tool merging 3 manual emissions test processes: reduced runtime from 30 min to 2-3 min, saving ~\$600k annually (tool still in production, 2021).
- Led piloted and chassis-dyno testing/calibration for Euro-6 diesel, SCR, and hybrid engines (Cummins, John Deere, Mahindra).

#### **Graduate Research Assistant – Digital Twins & Vehicle Design**

Clemson University (Army GVSC Collaboration), Clemson, SC

Aug 2025 – Present

- Collaborated with U.S. Army GVSC on digital twin integration for ground vehicle systems.
- Evaluated 5 non-functional requirements (-ilities) for tracked military vehicles; delivered formal definitions and mathematical models (Maneuverability, Trafficability, etc.).

#### **Research Associate – Robotics & Automation**

Club First Robotics (Robotic Company), Jaipur, India

Jul 2016 – Nov 2017

- Designed and validated Instant Water Purification & Vending System (IWPVS) for rural deployment.
- Built service robot mechanical structure, validated kinematics and dynamics via FEA.

#### **Engineering Intern (Automotive Engineering)**

Honda Cars India Ltd. (Car Manufacturer), Greater Noida, India

Jan 2016 – Jun 2016

- Designed "Master Jig" eliminating petrol/diesel changeover (38.5s loss to 0s); reduced operator count and advanced to supplier prototype.

### EDUCATION

M.S., Mechanical Engineering (Thesis)

Clemson University, Clemson, SC

Aug 2023 – Dec 2025 (Expected)

GPA: 3.85 | Thesis: AI-Augmented Framework for Bio-Inspired Design (FFP Matching + LLM Replication)

- Investigated the type of analogies adopted in bio-inspired design.

- Designed and iteratively optimized a 10-step LLM-assisted analysis pipeline (GPTDesign\_Analysis) using human feedback to improve functional decomposition and bio-inspired analogy matching; no model fine-tuning, only workflow and prompt engineering.

*B.Tech, Mechanical Engineering*

*JECRC University, Jaipur, India*

*Aug 2012 – Aug 2016*

Founder, JU Automotive Club | Led teams to national wins (Eco-Kart, Electric Baja) | Best Design Award

## PUBLICATIONS & PATENTS

*SAE WCX 2026 (Accepted)*

- Effects of Targeted Instruction on Non-Functional Requirement Understanding
- Enhancing Track Vehicle Design Engineering Through Digital Twins

*ASME IDETC-CIE 2025 (Accepted Poster)*

- Identifying Adopted Strategies in Bio-Inspired Design via Critical Chain and FFP Matching

*Patent Pending*

- Self-Powered Whale-Bot Tagging Device (Non-invasive, kinetic energy harvesting)

## SELECTED PROJECTS

- *Whale-Bot Tagging Device (2024–2025)*: Bio-inspired marine tracking tag; harvests whale motion for long-duration power; patent filing in process (supervised by Dr. Cameron Turner).
- *ECU IP-Theft Risk Study (2023–2024, Georgia Tech Collab)*: Demonstrated ECU benchmarking vulnerabilities; presented to cybersecurity faculty (Dr. Saman Zonouz and team).
- *Li-ion Battery On-Board Diagnostics (2023–2024)*: EKF-based state estimation with hybrid thermal-electrical modeling in MATLAB/Simulink.
- *NURBS/Bezier Design Automation Tool*: Co-developed a modular C++/Python program that reads Excel-based control nets to generate and visualize Bezier, B-spline, and NURBS curves and surfaces. Implemented knot insertion, degree elevation/reduction, and surface-construction routines (bilinear, general cylinder, revolution) to support automated geometric design.

## TECHNICAL SKILLS

*CAD & Prototyping*: SolidWorks, Fusion 360, Inventor, ANSYS | 3D Printing (FDM/SLA)

*Programming*: Python, MATLAB, Simulink, C++, R

*Testing & Diagnostics*: DOE, FEA formulation, Engine Dynos, Climate Chambers, Emissions Analyzers, DFMEA, Functional Modeling, Geometric Dimensioning & Tolerancing (GD&T), Critical Chain Modeling

*Tools & Systems*: ETAS INCA, Cummins Calterm, Horiba MEXA, SCADA, PLC

*Frameworks*: DFMEA, Functional Modeling, Tolerance Analysis, Critical Chain Modeling

## PROFESSIONAL AFFILIATIONS & CERTIFICATIONS

*Research*: CEDAR Lab, Deep Orange, VIPR-GS (Army Autonomy Prototyping)

*Certificate*: Professional English-Speaking course by Georgia Institute of Technology via Coursera, Nov 2025 (ID: ABH7056WXEG5; URL: <https://coursera.org/verify/ABH7056WXEG5>)